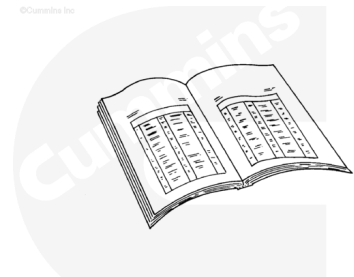


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

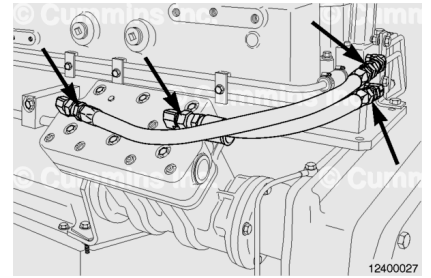
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : Not all engines are equipped with this option.

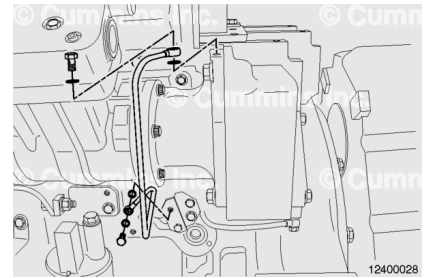
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Drain the engine cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html>)
- Remove the air inlet and outlet connections from the air compressor.

Remove

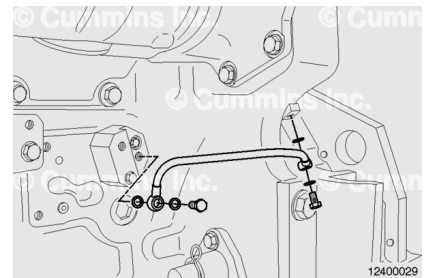
Remove the air compressor inlet and outlet coolant connections.



Remove the oil supply tube from the air compressor drive and the lubricating oil connection block by removing the two securing banjo bolts and washers from the oil line.

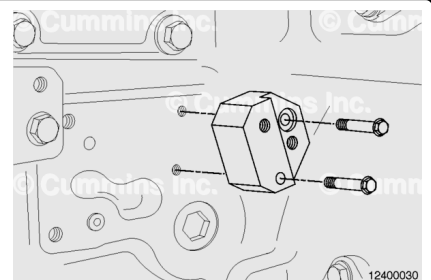


Remove the oil supply tube from the air compressor idler and the lubricating oil connection block by removing the two banjo bolts and washers from the oil line.



Remove the two capscrews securing the air compressor lubricating oil supply block.

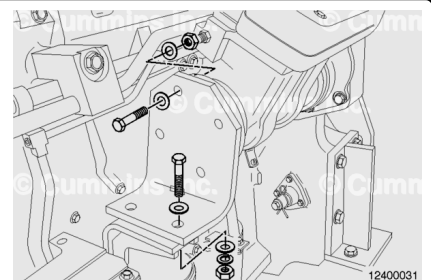
Remove the air compressor lubricating oil supply block and discard the o-ring.



Remove the six capscrews securing the rear air compressor support bracket.

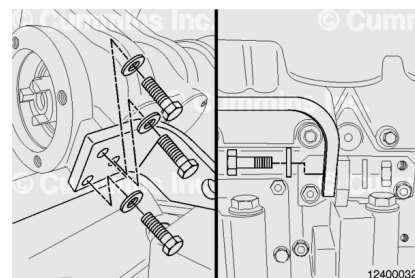
Remove the rear air compressor support bracket.

Discard the gasket.



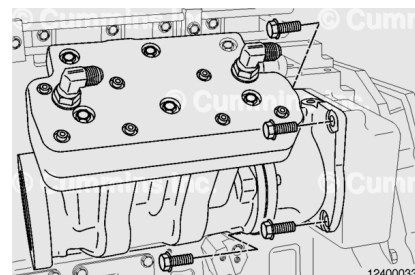
Remove the four capscrews securing the center air compressor support bracket.

Remove the center air compressor support bracket.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

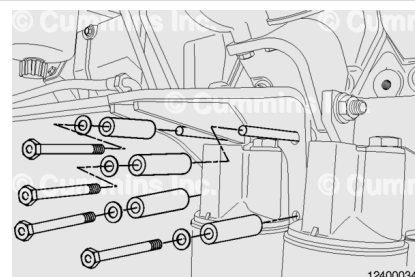


Remove the four mounting capscrews securing the air compressor and air compressor drive to the flywheel housing.

Use care **not** to damage the drive gear teeth. Remove the air compressor and drive assembly from the flywheel housing.

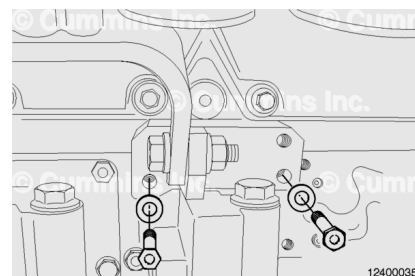
Loosen the four capscrews and spacers securing the rear air compressor lower support bracket to the lubricating filter head, if required, to remove the bracket.

Remove the rear air compressor lower support bracket.



Loosen the two capscrews securing the center air compressor lower support bracket to the lubricating filter head, if required, to remove the bracket.

Remove the center air compressor lower support bracket.

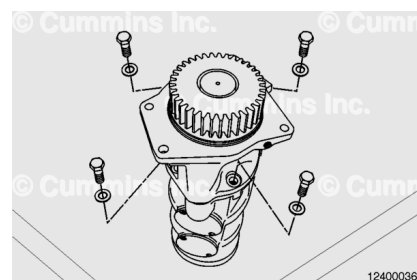


Disassemble

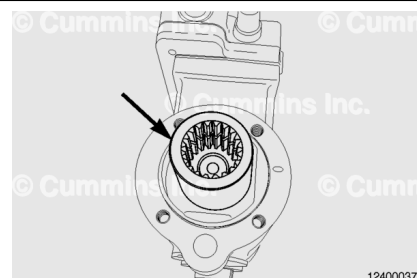
With the air compressor and drive assembly resting on a flat surface, rear end down, remove the four capscrews securing the air compressor to the drive unit.

Remove the drive unit from the air compressor.

Discard the gasket.



Remove the coupling from the air compressor spline shaft.



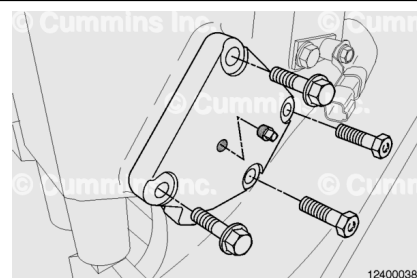
Rear Gear Train

Remove the square plug from the air compressor idler gear mounting plate.

Remove the four mounting capscrews securing the air compressor idler gear plate to the flywheel housing.

Remove the air compressor idler gear plate from the flywheel housing.

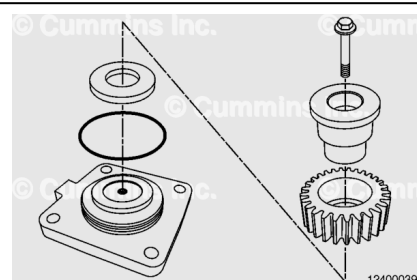
Discard the gasket.



Remove the bolt securing the idler gear shaft onto the air compressor idler gear plate.

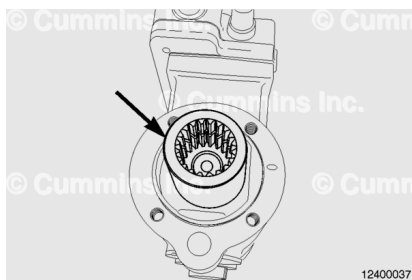
Disassemble the idler shaft, gear, and thrust plate.

Discard the o-ring.



Assemble

Install the coupling onto the spline shaft of the air compressor.



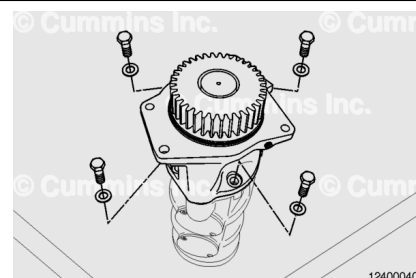
With the air compressor resting on a flat surface, rear end side down, install the drive unit to the air compressor.

Install a new gasket.

The coupling **must** be aligned to the spline shaft of the drive unit.

Install the four mounting capscrews securing the air compressor to the drive unit.

Torque Value: 80 n•m [59 ft-lb]



Rear Gear Train

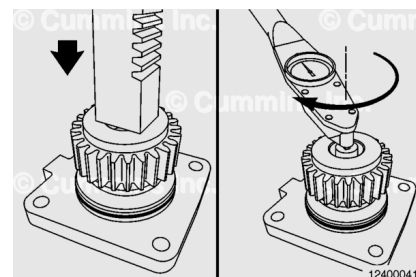
Install a new o-ring.

Assemble the air compressor idler gear mounting plate, thrust plate, gear, and idler shaft.

Press fit the assembly.

Tighten the mounting capscrews securing the idler gear shaft to the mounting plate.

Torque Value: 290 n•m [214 ft-lb]



Install the square plug into the air compressor idler gear mounting plate.

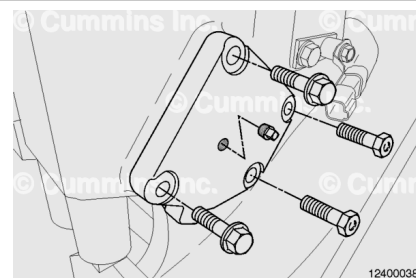
Install a new gasket.

Install the air compressor idler gear plate to the flywheel housing.

The idler gear **must** properly engage with the cam gear.

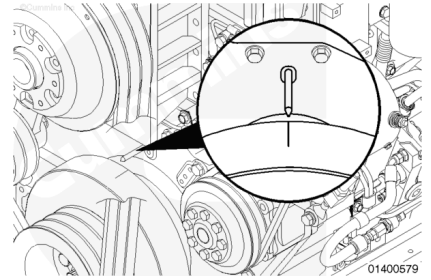
Install the four mounting capscrews.

Torque Value: 108 n•m [80 ft-lb]



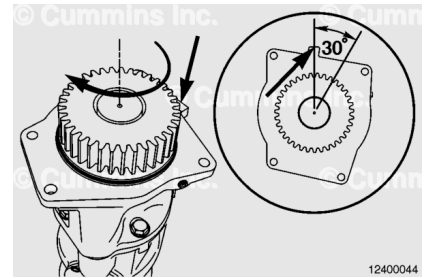
Install

Bar the engine to align the 1.6TOP indicator on the damper with the pointer.



Rotate the air compressor drive gear by hand. Feel for the compression peak point.

Once the peak point is identified, rotate the air compressor drive gear approximately 30 degrees **clockwise** from the peak point (as viewed from the drive gear side).

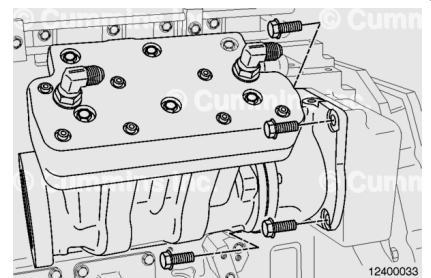


Install a new gasket. Insert the air compressor and drive unit assembly into the power take-off port in the flywheel housing.

The air compressor drive gear must properly mesh with the air compressor idler gear.

Install the top two mounting capscrews and washers.

Torque Value: 68 n•m [50 ft-lb]

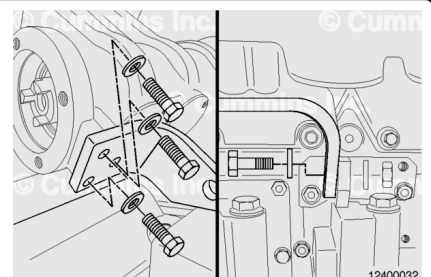


Assemble the air compressor center support bracket to the air compressor.

Install the three mounting capscrews to the air compressor support bracket.

Finger tighten the capscrews.

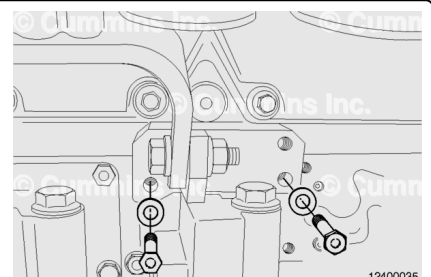
Do **not** torque the capscrews at this time.



Assemble the two mounting capscrews securing the air compressor center lower support bracket to the filter head bracket.

Finger tighten the capscrews.

Do **not** torque the capscrews at this time.

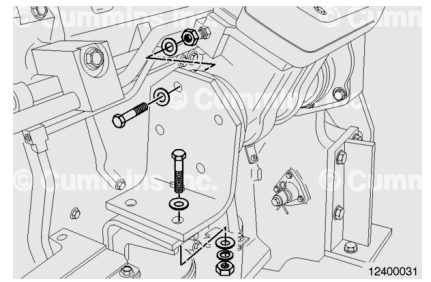


Install a new gasket. Install the air compressor rear housing support plate.

Install the four mounting capscrews securing the air compressor rear housing support bracket to the air compressor.

Finger tighten the capscrews.

Do **not** torque the capscrews at this time.



Use the alignment dowels (1) to install the rear lower air compressor support bracket.

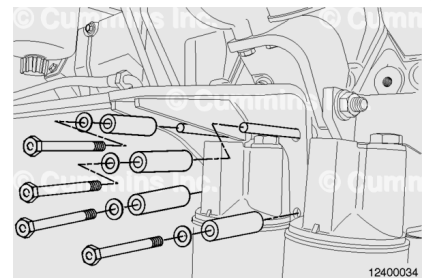
Install the lower two mounting capscrews and spacers.

Finger tighten the capscrews.

Remove the alignment dowels (1) and install the upper two mounting capscrews and spacers.

Finger tighten the capscrews.

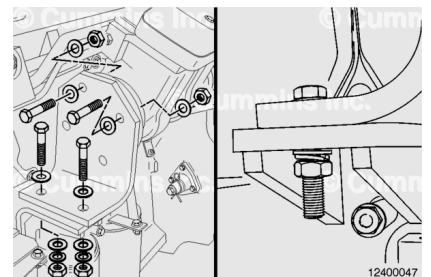
Do **not** torque the capscrews at this time.



Install the two capscrews and nuts securing the upper and lower rear air compressor support brackets.

Finger tighten the capscrews.

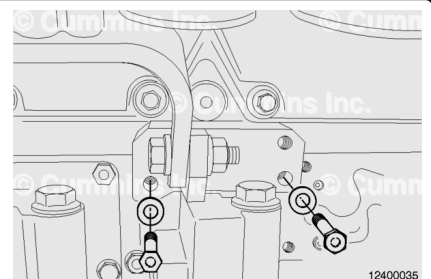
Do **not** torque the capscrews at this time.



Install the capscrew and nut securing the upper and lower center air compressor support brackets.

Finger tighten the capscrew and nut.

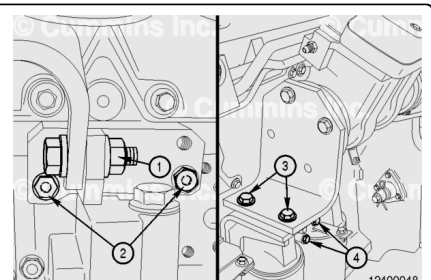
Do **not** torque at this time.



All slack in the mounting brackets **must** be taken up by finger tightening all capscrews.

Tighten the four capscrews securing the air compressor rear mounting plate.

Torque Value: 88 n•m [65 ft-lb]



Tighten the three capscrews securing the air compressor center support bracket.

Torque Value: 50 N·m [37 ft-lb]

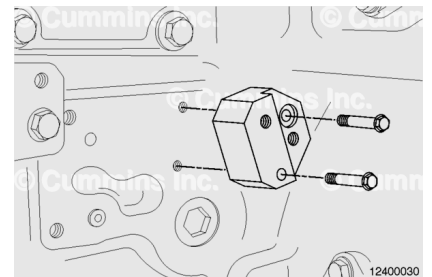
Tighten the remaining center and rear support brackets in the following order.

Torque Value:

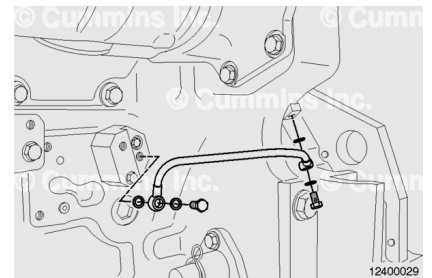
1. 170 N·m 125 ft-lb
2. 68 N·m 50 ft-lb
3. 68 N·m 50 ft-lb
4. 68 N·m 50 ft-lb

Install a new o-ring. Install the air compressor lubricating oil supply block to the rear left bank of the engine block.

Torque Value: 68 N·m [50 ft-lb]



Install the oil supply tube to the air compressor idler and the lubricating oil connection block by installing the two banjo bolts and washers to the oil line.



Install the lubricating oil supply from the air compressor lubricating oil supply block to the air compressor drive unit. Use two banjo bolts and washers.

Torque Value: 10 N·m [89 in-lb]

Use a p-clip to secure the lubricating oil supply tube.

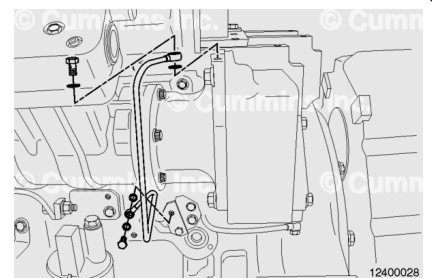
Install the clip to the drive unit with the mounting capscrew in location (1).

Torque Value: 68 N·m [50 ft-lb]

Install the lubricating oil supply tube from the air compressor lubricating oil supply block to the air compressor idler plate. Use two banjo bolts and washers.

Torque Value: 10 N·m [89 in-lb]

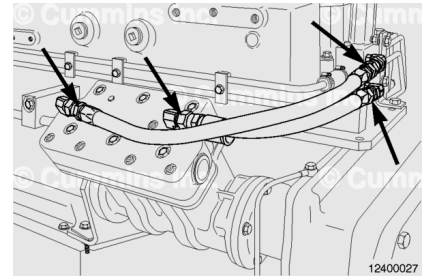
Use a p-clip to secure the lubricating oil supply tube.



Install the clip to the idler gear mounting plate with the mounting capscrew and the associated hardware in location (1).

Torque Value: 68 n•m [50 ft-lb]

Install the air compressor inlet and outlet coolant lines.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the air inlet and outlet connections to the air compressor.
- Fill the engine cooling system. Refer to Procedure 008-018 in Section 8.

(/qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html)

- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the engine and check for leaks.

Last Modified: 27-Jul-2009
